

Task 3: Hypothesis Testing

COURSE NAME

CSDS 413 Introduction to Data Analysis

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Context

Car sales people usually claim that Japanese cars have better mileage than US cars. As a critical thinker, you know not to immediately believe what somebody who is trying to sell you something says. However, as a data scientist, you are curious to see whether this claim has any statistical basis.

You found some data collected in “cars.csv”. The first column contains a sample of measured mileage per hour for different US cars and the second column contains a sample of measured mileage per hour for different Japanese Cars. Answer the following questions:

- (a) How many US and how many Japanese cars does the dataset contain?
- (b) State the hypothesis you are testing to scrutinize the sales person's claim.
- (c) What is the statistic you would compute to test this hypothesis and its theoretical distribution?
- (d) What is the value of your statistic and what is its p-value?
- (e) What is your conclusion about the sales person's claim?

Note: Be sure to check any conditions or assumptions of the chosen test before computing statistics and drawing your conclusion.

1 Part A

How many US and how many Japanese cars does the dataset contain?

From the context, "first column contains a sample of measured mileage per hour for different US cars and the second column contains a sample of measured mileage per hour for different Japanese Cars". This ought to mean that each row, regardless of if it contains a mileage that has been seen before, represents a different car for the purpose of counting. Below is a little Python REPL to count the number of rows for each column containing a mileage:

```
(assignment2-py3.12) jande@BT-7274:~/csds413_assignments/assignment2/
task3_hypothesis_testing/datasets$ python
Python 3.12.0 (main, Jun  9 2025, 09:52:12) [GCC 13.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import pandas as pd
>>> cars = pd.read_csv("cars.csv", header=None)
>>> us_cars = len(cars[cars[0].notna()])
>>> us_cars
249
>>> jp_cars = len(cars[cars[1].notna()])
>>> jp_cars
79
```

Figure 1: Python REPL counting the number of cars of each type in the dataset.

There are 249 US cars and 79 Japanese cars.

2 Part B

State the hypothesis you are testing to scrutinize the sales person's claim.

H_0 : The average mileage of Japanese cars is less than or equal to the average mileage of US cars.

H_1 : The average mileage of Japanese cars is greater than the average mileage of US cars.

3 Part C

What is the statistic you would compute to test this hypothesis and its theoretical distribution?

I'm going to reasonably assume the measurement of US car mileages have no dependency on Japanese car mileages and vice versa. That being said, the hypothesis in part B is a statement on the comparison of average mileage, which I framed in that manner because the claim from the problem context comments on what people usually claim about these cars, so a comparison of average mileage of these cars ought to be most appropriate.

With that established, choosing an appropriate test to compare the means of two independent groups immediately brings two-sample Student's t-test to mind. This test requires that the groups be independent, that the samples should be about Normal, and their variances should be about the same. Below is the function and corresponding Q-Q Plot I generated to see how closely each group resembles a Gaussian:

```
def main():
    data = pd.read_csv('../datasets/cars.csv', header=None)
    us_mileage = data[0].dropna().to_numpy()
    jp_mileage = data[1].dropna().to_numpy()

    sns.set_style("whitegrid")
    sns.set_palette("muted")
    _, axes = plt.subplots(1, 2, figsize=(12, 6))

    percentiles = np.linspace(0, 100, 75)
    norm_quantiles_us = np.percentile(np.random.normal(loc=np.mean(us_mileage),
    scale=np.std(us_mileage), size=10_000), percentiles)
    sample_quantiles_us = np.percentile(us_mileage, percentiles)

    norm_quantiles_jp = np.percentile(np.random.normal(loc=np.mean(jp_mileage),
    scale=np.std(jp_mileage), size=10_000), percentiles)
    sample_quantiles_jp = np.percentile(jp_mileage, percentiles)

    axes[0].scatter(norm_quantiles_us, sample_quantiles_us, alpha=0.5,
                     s=20, color='blue', edgecolor='blue')
    axes[0].plot([min(norm_quantiles_us), max(norm_quantiles_us)],
                 [min(norm_quantiles_us), max(norm_quantiles_us)],
                 'k--', lw=2, alpha=0.7)
    axes[0].set_title("Q-Q Plot: US Cars Mileage", fontsize=14)
    axes[0].set_xlabel("Theoretical Quantiles", fontsize=12)
    axes[0].set_ylabel("Sample Quantiles", fontsize=12)

    axes[1].scatter(norm_quantiles_jp, sample_quantiles_jp, alpha=0.5,
                     s=20, color='purple', edgecolor='purple')
    axes[1].plot([min(norm_quantiles_jp), max(norm_quantiles_jp)],
                 [min(norm_quantiles_jp), max(norm_quantiles_jp)],
                 'k--', lw=2, alpha=0.7)
    axes[1].set_title("Q-Q Plot: Japanese Cars Mileage", fontsize=14)
    axes[1].set_xlabel("Theoretical Quantiles", fontsize=12)
    axes[1].set_ylabel("Sample Quantiles", fontsize=12)

    plt.tight_layout()
    plt.savefig(f"../figures/qqplot.png", bbox_inches='tight', dpi=300)
```

Figure 2: Function for plotting theoretical against sample quantiles for US and Japanese cars.

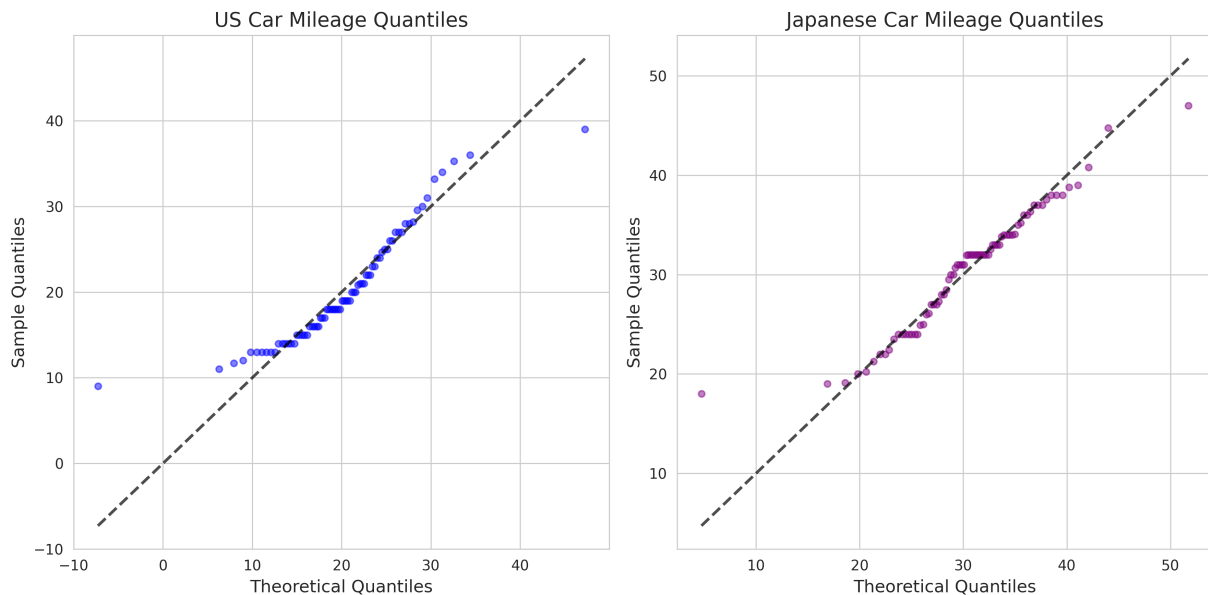


Figure 3: Q-Q plots for US and Japanese cars quantiles.

The quantiles for both the US cars and the Japanese cars stay close to the agreement line through the Q-Q Plot, especially toward the middle quantiles, though they do both have a couple that visually deviate quite far from the agreement line in the more extreme cases. The plot seems to suggest they are both approximately Normal, but it would be helpful to also visualize the distribution shape directly as well, not to mention they could be plotted on the same figure to give a general idea of which country may be produce higher mileage cars in the average-case:

```
def main():
    data = pd.read_csv('../datasets/cars.csv', header=None)
    us_mileage = data[0].dropna()
    jp_mileage = data[1].dropna()

    sns.set_style("whitegrid")
    sns.set_palette("muted")

    plt.figure(figsize=(10, 6))

    sns.histplot(us_mileage, bins=30, kde=False, color='blue', edgecolor='white',
alpha=0.6, label='US Cars')
    sns.histplot(jp_mileage, bins=30, kde=False, color='red', edgecolor='white',
alpha=0.6, label='Japanese Cars')

    plt.title('Histogram: US vs Japanese Cars Mileage', fontsize=16)
    plt.xlabel('Mileage', fontsize=14)
    plt.ylabel('Count', fontsize=14)
    plt.legend(fontsize=12)

    plt.tight_layout()
    os.makedirs("../figures", exist_ok=True)
    plt.savefig("../figures/histogram.png", bbox_inches='tight', dpi=300)
```

Figure 4: Function for plotting US and Japanese car mileage distributions.

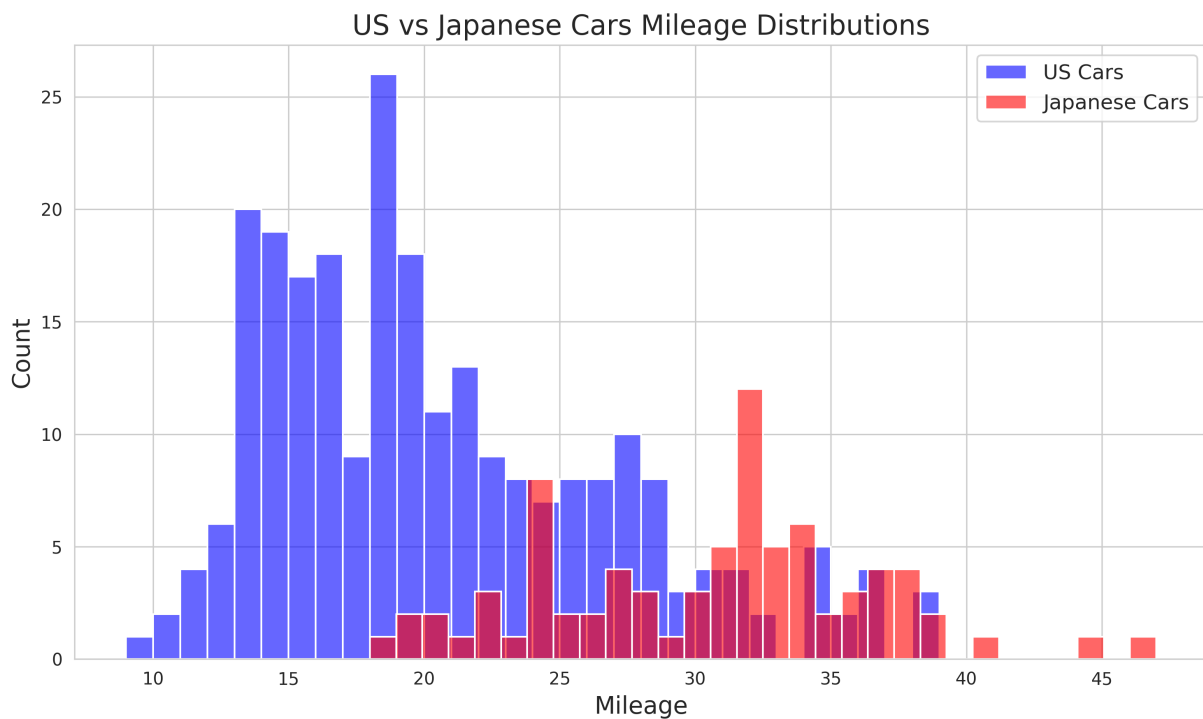


Figure 5: Histogram plots for US and Japanese cars mileages.

This result gives us pretty decent visual confirmation that these groups both approximate Normal distributions, and the addition histogram plot does well to visualize the peculiarities in the distribution mass that attribute to the deviations from the agreement line that we saw in the Q-Q Plot. For example, the heavier left tail of the US car distribution corresponding to the deviations above the agreement line or the right tail tapering off a bit faster than the right in the case of Japanese car mileages corresponding to the deviations below the agreement line. Also like I mentioned, the histogram gives us a rough confirmation that the average Japanese car mileage is higher than the US cars.

We need to also confirm that the variance are about the same, so for this purpose I produced box plots corresponding to each country:

```
def main():
    data = pd.read_csv('../datasets/cars.csv', header=None)
    us_mileage = data[0].dropna()
    jp_mileage = data[1].dropna()

    df = pd.DataFrame({
        'Mileage': pd.concat([us_mileage, jp_mileage], ignore_index=True),
        'Region': ['US'] * len(us_mileage) + ['Japan'] * len(jp_mileage)
    })

    sns.set_style("whitegrid")
    sns.set_palette("muted")

    plt.figure(figsize=(8, 6))
    ax = sns.boxplot(x='Region', y='Mileage', data=df, showfliers=True)

    ax.set_title('Car Mileages for US vs Japanese Cars', fontsize=16)
    ax.set_xlabel('')
    ax.set_ylabel('Mileage', fontsize=12)

    plt.tight_layout()
    plt.savefig("../figures/boxplot.png", bbox_inches='tight', dpi=300)
```

Figure 6: Function for plotting US and Japanese car mileages as box plots.

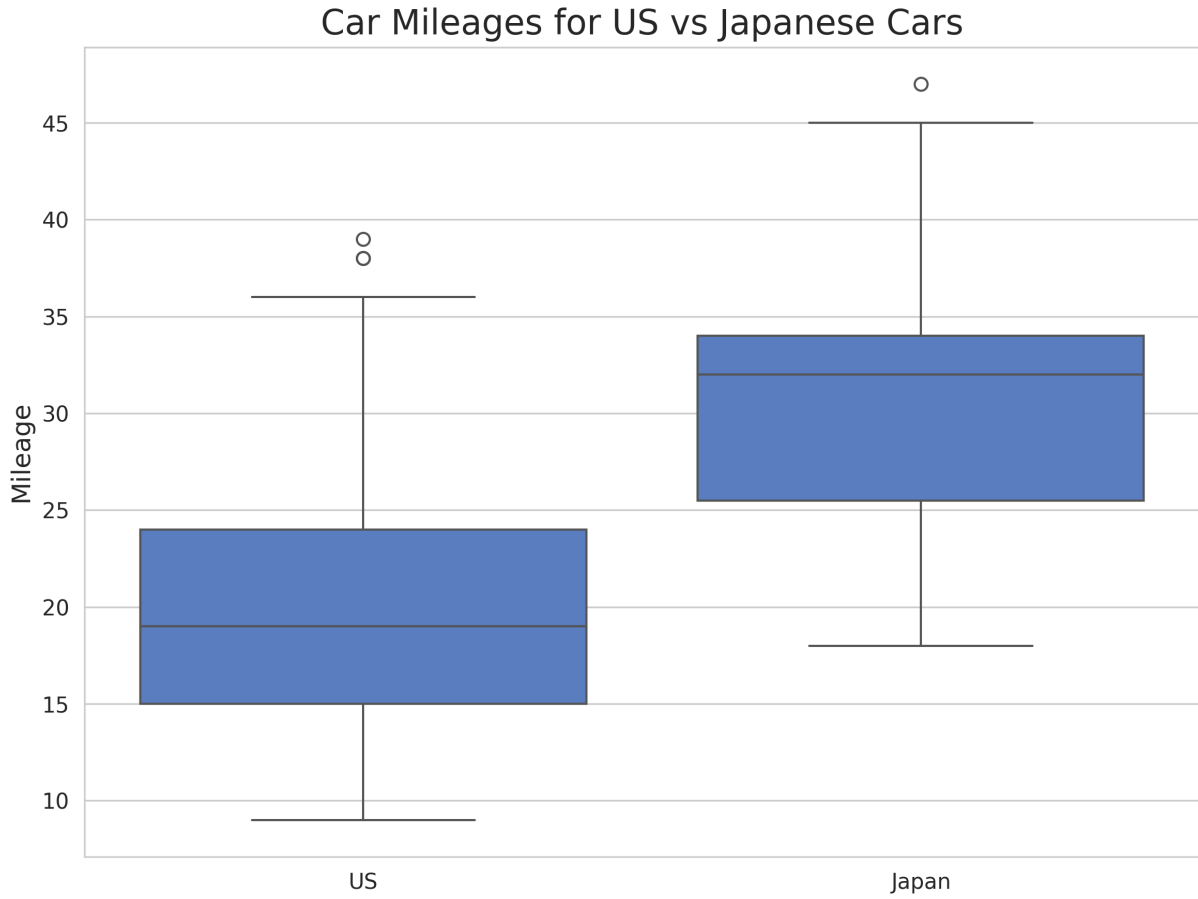


Figure 7: Box plots for US and Japanese cars mileages.

The box plots show a very promising signal that their inter-quartile ranges are about the same size and it shows that their whiskers protrude by about the same amount both above and below their IQR. We can also see a comparison of the medians from this visual, and given that the respective distributions are sufficiently symmetrical, then we have another rough signal that the Japanese cars may greater mileage on average.

That being said, these visualizations confirm that the underlying conditions of the t-test have been adequately met. This test statistic follows a t-distribution, which models the sampling distribution of the standardized difference between sample means. It is similar to the Normal distribution but with heavier tails, especially for smaller sample sizes, reflecting an increased uncertainty. In this case, the t-distribution can be expressed as:

$$t = \frac{\bar{x}_{jp} - \bar{x}_{us}}{\sigma_{\bar{x}_{jp} - \bar{x}_{us}}}$$

where $\sigma_{\bar{x}_{jp} - \bar{x}_{us}}$ is the standard error of the difference between sample means:

$$\sigma_{\bar{x}_{jp} - \bar{x}_{us}} = \sqrt{\frac{\sigma_{jp}^2}{n_{jp}} + \frac{\sigma_{us}^2}{n_{us}}}$$

This test statistic follows this distribution with $n_{jp} + n_{us} - 2$ degrees of freedom under H_0 . Because H_1 is one-sided (Japanese > US), I will perform a right-tailed test, rejecting H_0 if the t-statistic is sufficiently large.

4 Part D

What is the value of your statistic and what is its p-value?

```
def main():
    data = pd.read_csv('../datasets/cars.csv', header=None)

    us_mileage = data[0].dropna().to_numpy()
    jp_mileage = data[1].dropna().to_numpy()

    n_us = len(us_mileage)
    n_jp = len(jp_mileage)

    mean_us = np.mean(us_mileage)
    mean_jp = np.mean(jp_mileage)

    var_us = np.var(us_mileage, ddof=1)
    var_jp = np.var(jp_mileage, ddof=1)
    se = np.sqrt(var_us/n_us + var_jp/n_jp)

    t_stat = (mean_jp - mean_us) / se
    df = n_us + n_jp - 2
    p_value = stats.t.sf(t_stat, df)

    print(f"t-statistic: {t_stat:.4f}")
    print(f"p-value: {p_value:.4e}")
```

Figure 8: Function for computing the t statistic and its p-value.

The t-statistic was reported by the function as 12.946, with a p-value of 1.6197×10^{-31} .

5 Part E

What is your conclusion about the sales person's claim?

The t-statistic of 12.946 indicates that the observed difference in mean is nearly 13 standard errors away from zero. Pairing this result with a p-value of 1.6197×10^{-31} suggests that if the null hypothesis were true, the probability of observing a t-statistic this high would be virtually zero. This provides overwhelming evidence in favor of rejecting the null hypothesis and concluding that the salesperson is correct in claiming Japanese car mileage is better than US car mileage.